Aim : Introduction to Truffle, Establish local blockchain using Truffle and deployment of Smart Contract

Objectives :

1. To create and deploy smart contracts using Remix IDE.
2. To use Ganache as a local blockchain for testing smart contracts.
3. To record product details on the blockchain.
4. To track product movement between different participants in the supply chain.
5. To verify transactions and stored product information.

Theory :

1. Blockchain

Blockchain is a distributed digital ledger that stores transactions in blocks. These blocks are linked together using cryptographic hashes. Once a transaction is recorded, it is difficult to modify, which helps maintain data integrity and transparency.

2. Smart Contract

A smart contract is a program stored on a blockchain that executes predefined instructions when specified conditions are met. Smart contracts are commonly written in Solidity for Ethereum-compatible blockchains. They can be used for applications such as supply-chain management, voting systems, and digital transactions.

3. Truffle

Truffle is a development framework used to develop, compile, test, and deploy smart contracts. It provides tools that simplify the smart contract development process. It can also be used to manage contract artifacts and deployment scripts.

4. Ganache

Ganache is a local blockchain development tool used to simulate an Ethereum blockchain on a computer. It provides test accounts with sample cryptocurrency, allows transactions to be executed locally, and displays transaction details such as transaction hashes, gas usage, and block information. It is useful for testing smart contracts without using real cryptocurrency.

5. MetaMask

MetaMask is a cryptocurrency wallet that allows users to connect to blockchain networks and interact with decentralised applications. In this experiment, MetaMask is connected to the local Ganache blockchain. A Ganache test account is used to deploy and interact with smart contracts.

6. Remix IDE

Remix IDE is a browser-based development environment for writing, compiling, debugging, and deploying Solidity smart contracts. It provides a Solidity compiler and a deployment interface. By selecting the *Injected Provider – MetaMask* environment, Remix can use MetaMask to connect to the selected blockchain network.

7. Smart Contract Deployment

Deployment is the process of publishing a compiled smart contract to a blockchain. During deployment, a transaction is submitted from an account. The deployed contract receives a unique address, which is used to interact with it later. On Ganache, transactions can be inspected locally without spending real cryptocurrency.

8. Supply-Chain Management Using Smart Contracts

A supply chain involves the movement of products from producers to customers through intermediaries. In this experiment, separate smart contracts represent the Farmer, Distributor, Retailer, and Customer. The farmer records product details, the distributor receives the batch, the retailer receives it from the distributor, and the customer records the purchase. This demonstrates how blockchain can maintain a traceable record of product movement.

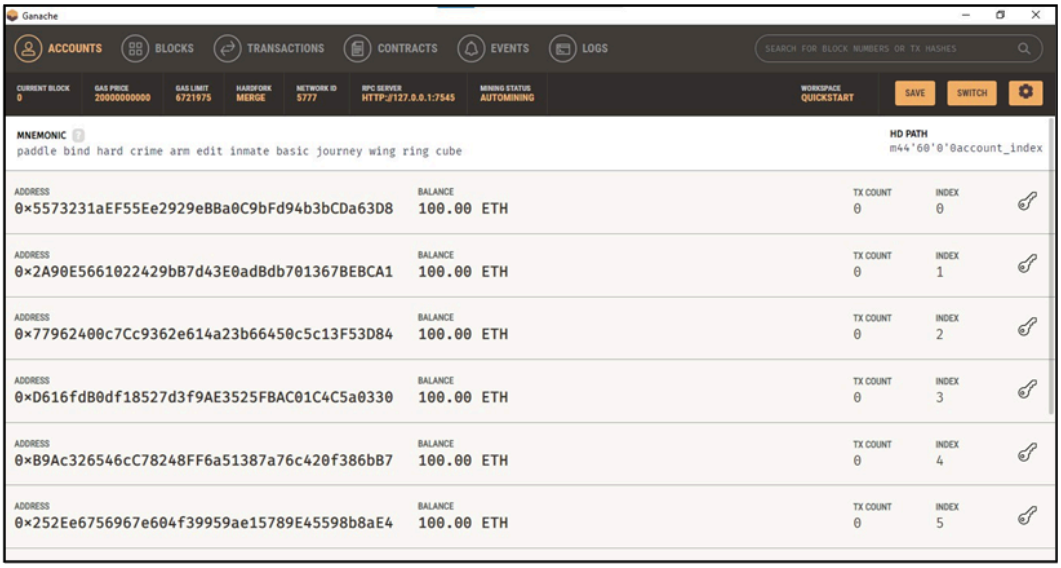
Working Principle

First, Ganache is started to create a local blockchain. MetaMask is connected to the Ganache network, and Remix IDE is connected through MetaMask. The Solidity contracts are compiled and deployed in the correct order. The deployed contracts are then tested by recording a product batch and transferring its details through the distributor, retailer, and customer stages. The transactions are verified using Ganache.

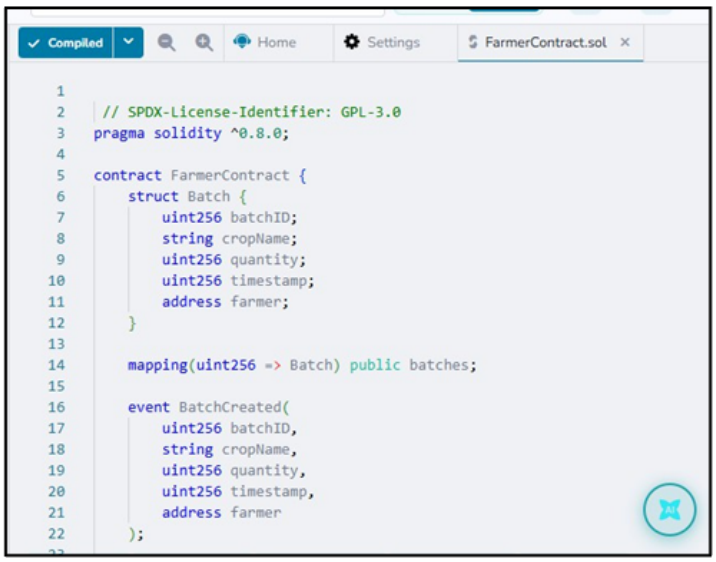
Advantages

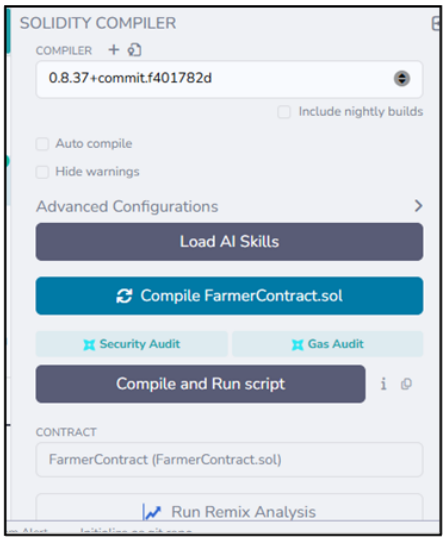
- Provides a local environment for testing blockchain applications.
- Uses test accounts, so real cryptocurrency is not required.
- Displays transactions and block details for verification.
- Demonstrates traceability and transparency in supply-chain operations.
- Helps developers identify errors before deploying contracts to a live network.

Implementation :



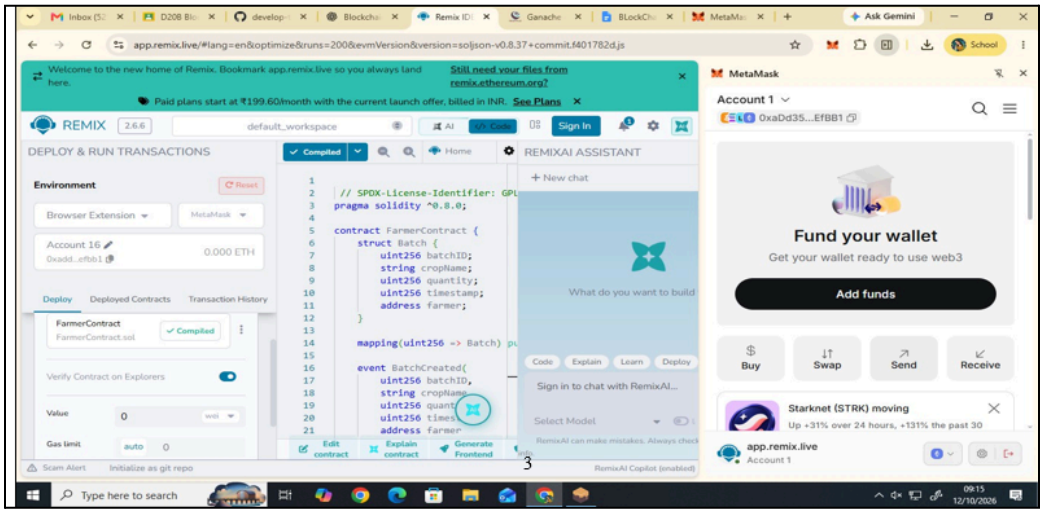
Start Ganache

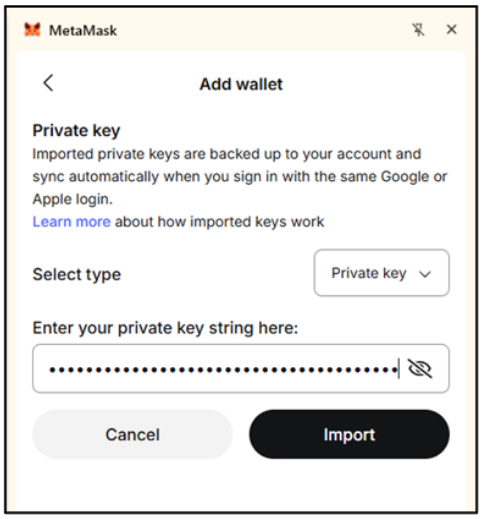




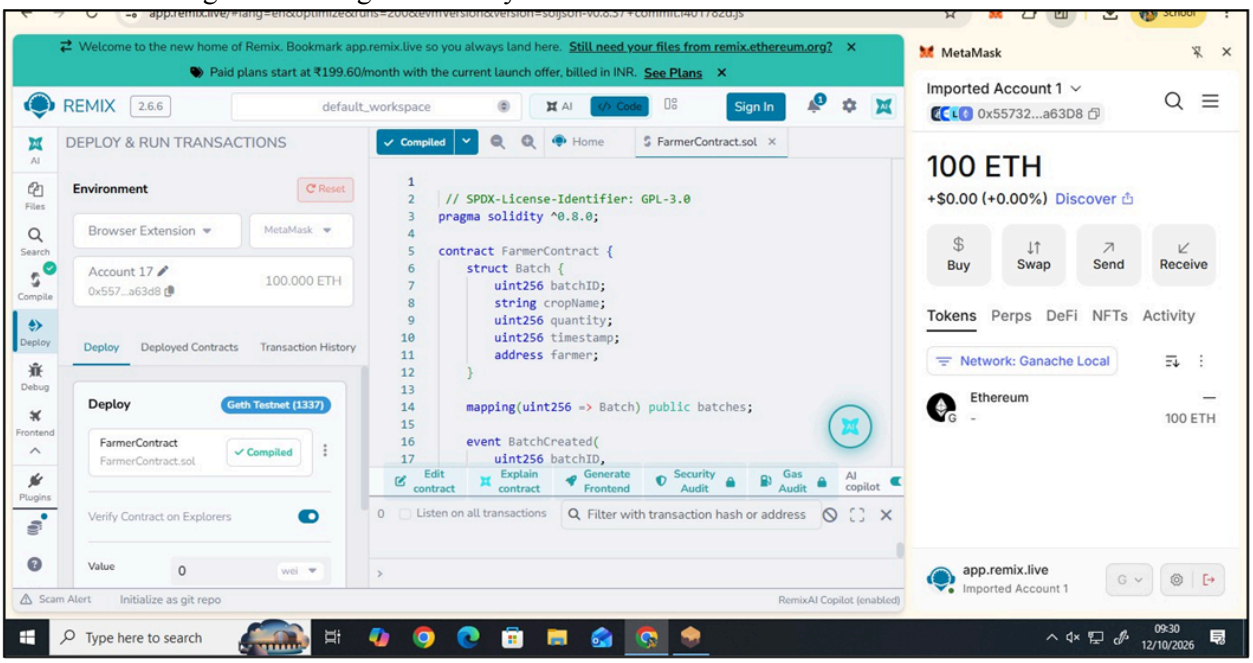
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FarmerContract.sol

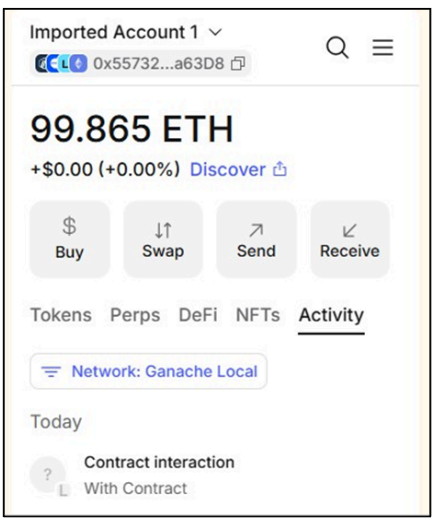


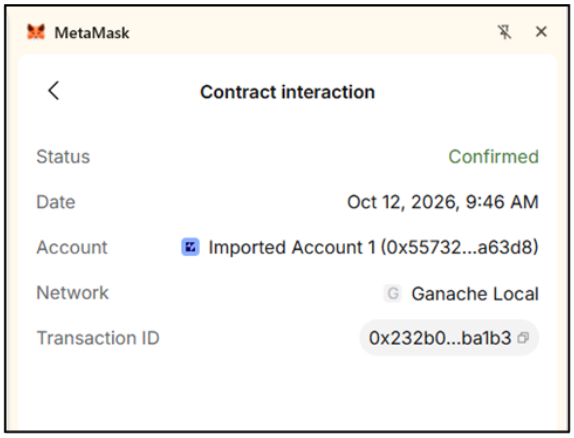


Metatask - Adding Wallet using Private Key



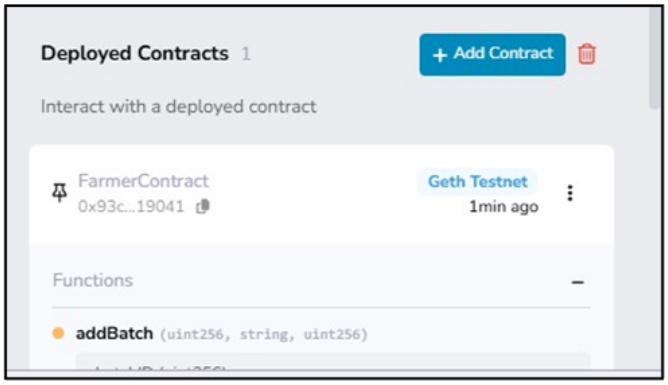
Deployed FarmerContract.sol







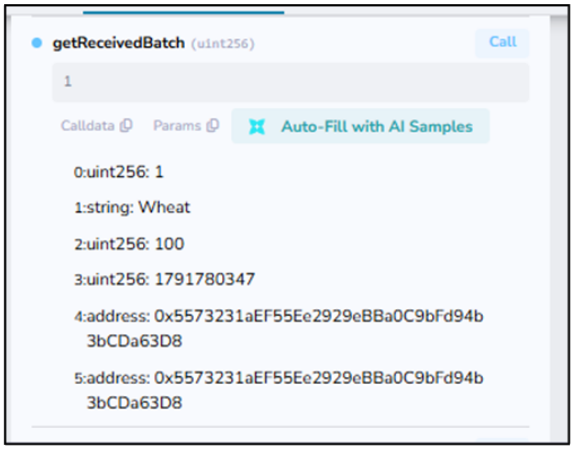
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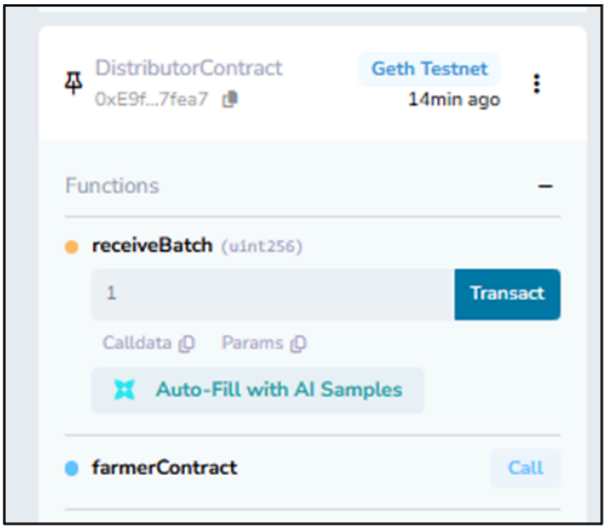


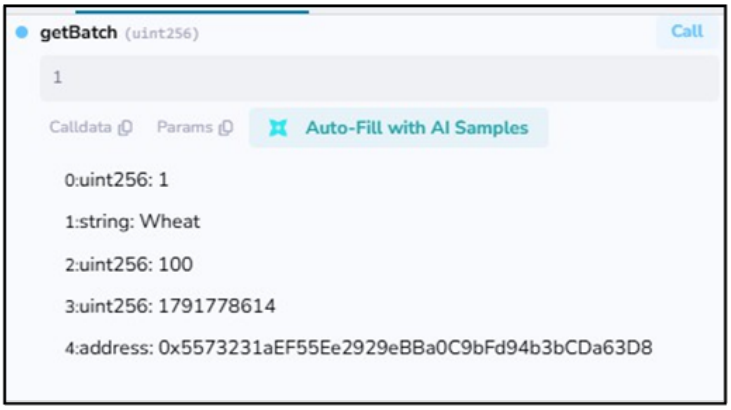


DistributedContract.sol



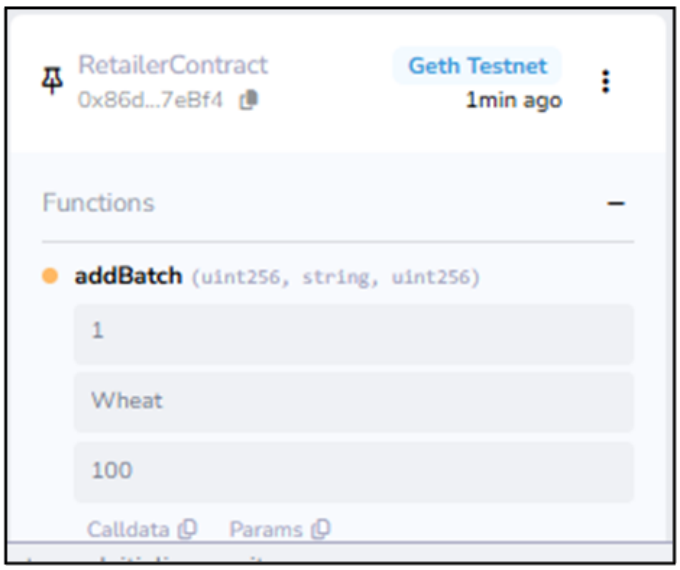






RetailerContract.sol

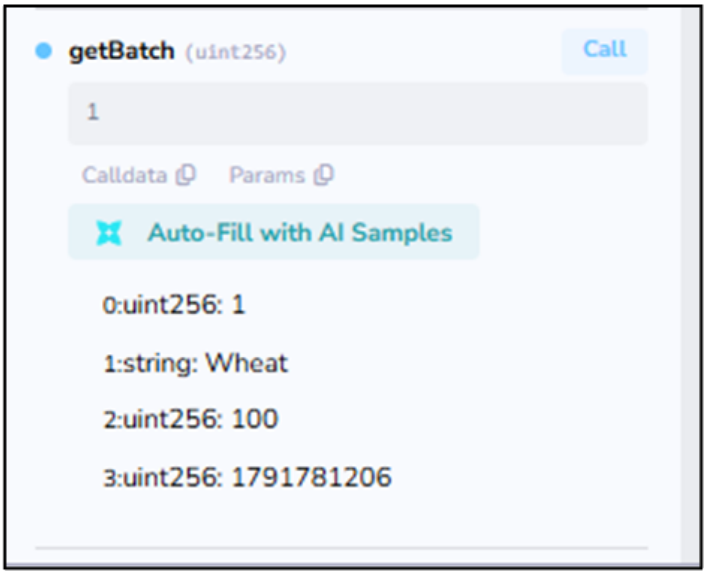


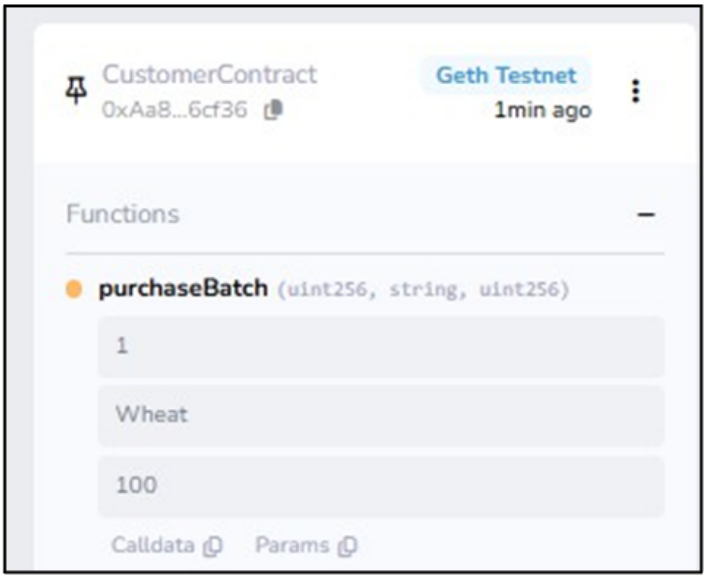




CustomerContract.sol







Output / Observations :

The smart contracts were tested successfully using Ganache and Remix IDE. The following observations were recorded:

1. FarmerContract stored and returned batch details.
2. Batch ID: 1
3. Product Name: Wheat
4. Quantity: 100
5. DistributorContract received batch 1 and returned the received batch details.
6. Contract deployment and function calls were tested through Remix IDE.

Screenshots of contract deployment, transaction execution, and returned values were recorded as evidence.

Result :

Thus, a blockchain-based supply chain management system was implemented and tested using Solidity smart contracts, Remix IDE, and Ganache. Product batch information was stored and verified, demonstrating the use of blockchain for supply chain traceability.

Conclusion :

The experiment demonstrated how smart contracts can be used to record and track products at different stages of a supply chain. Ganache provided a local blockchain environment for testing, while Remix IDE enabled contract compilation, deployment, and interaction.